

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 3 Winter 2025 (Supply.)

Course : Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTBS301 - ENGINEERING MATHEMATICS - III

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1 Laplace Transform of function $f(t)$ is defines as

(a) $\int_0^{\infty} \frac{1}{f(t)e^{-st}} dt$ (b) $\int_0^{\infty} f(t)e^{-st} dt$ (c) $\int_0^{\infty} f(t)e^{st} dt$ (d) $\int_0^{\infty} \frac{1}{f(t)e^{-st}} dt$

2 Laplace transform $f(t) = t \sin 3t$ is

(a) $\frac{3}{s^2+9}$ (b) $\frac{3s}{(s^2+3)^2}$ (c) $\frac{6s}{(s^2+3)^2}$ (d) $\frac{6s}{(s^2+9)^2}$

3 Inverse Laplace transform of $\frac{1}{(s+1)^2}$ is

(a) te^t (b) $-te^t$ (c) t^2e^{-t} (d) te^{-t}

4 Inverse Laplace transform of $\frac{1}{(s^2+64)}$ is

(a) $\cos 8t$ (b) $\cosh 8t$ (c) $\sin 8t$ (d) $\frac{\sin 8t}{8}$

5 Laplace transform of t^2 is

(a) $\frac{3}{s^2}$ (b) $\frac{2!}{s^2}$ (c) $\frac{4!}{s^2}$ (d) $\frac{1}{s^2}$

6 The Fourier cosine Transform of the function $f(t)$ is

(a) $F_c(s) = \int_0^{\infty} f(t) \cos st dt$ (b) $F_c(s) = \int_0^{\infty} f(st) \cos t dt$

(c) $F_c(s) = \int_0^{\infty} f(t) \cos t dt$ (d) None of These

7 Fourier Transformation is a linear combination, is

(a) True (b) False (c) None of these

8 The solution of $\frac{\partial^2 z}{\partial y^2} = \sin(xy)$ is

(a) $z = -x^2 \sin xy + yf(x) + \phi(x)$ (b) $z = -x^2 \cos xy + yf(x) + \phi(x)$

(c) $z = -y^2 \sin xy + yf(x) + \phi(x)$ (d) None of these

- 9 By eliminating a and b from $z = a(x + y) + b$, the partial differential equation formed is
 (a) $\frac{\partial z}{\partial x} = \frac{\partial z}{\partial y}$ (b) $\frac{\partial z}{\partial x} = \frac{\partial^2 z}{\partial y^2}$ (c) $\frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial y^2}$ (d) None of these
- 10 Which out of the following is an analytic function
 (a) $f(z) = \sin z$ (b) $f(z) = \operatorname{Im}(z)$ (c) $f(z) = \bar{z}$ (d) None of these
- 11 The analytic function is also a
 (a) Holomorphic function (b) Singular Point
 (c) Non-singular Point (d) None of these
- 12 If $f(z)$ is an analytic function with constant modulus, then $f(z)$ is a
 (a) Constant function (b) Harmonic function
 (c) Orthogonal (d) None of these
- Q2. Solve the following.
- A) Find the Laplace Transform of $L\{\cos t \cos 2t \cos 3t\}$ 6
- B) Find the Laplace Transform of $L\{te^{-4t} \sin 3t\}$. 6
- Q3. Solve the following.
- A) Find the inverse Laplace transform of $\frac{s^2 + 10s + 13}{(s-1)(s^2 - 5s + 6)}$. 6
- B) Use Laplace transform to solve $\frac{d^2 y}{dx^2} + 4\frac{dy}{dx} + 8y = 1$ where $y(0) = 0, y'(0) = 1$. 6
- Q4. Solve Any Two of the following. <https://www.batuonline.com>
- A) Using Fourier Integral show that $\int_0^\infty \frac{\omega \sin x \omega}{1 + \omega^2} = \frac{\pi}{2} e^{-x} (x > 0)$. 6
- B) Find the Fourier Transform of $f(x) = \begin{cases} (1 - x^2) & |x| \leq 1 \\ 0 & |x| > 1 \end{cases}$, and hence evaluate $\int_0^\infty \left(\frac{x \cos x - \sin x}{x^3} \right) \cos \frac{\pi}{2} dx$. 6
- C) Using Parseval's Identity show that $\int_0^\infty \frac{dt}{(t^2 + 1)^2} = \frac{\pi}{4}$. 6
- Q5. Solve Any Two of the following.
- A) Form the partial differential equation by eliminating function
 $z = f(x + it) + g(x - it)$. 6
- B) Solve $pz - qz = z^2 + (x + y)^2$. 6
- C) Solve the following equation by method of separation of variables
 $\frac{\partial^2 u}{\partial x \partial t} = e^{-t} \cos x$ given that $u = 0$ when $t = 0$ and $\frac{\partial u}{\partial t} = 0$ when $x = 0$. 6
- Q6. Solve Any Two of the following.

- A) Prove that $f(z) = \sin hz$ is analytic function and find its derivative. 6
- B) If $v = 3x^2y - y^3$, find its harmonic conjugate u . find $f(z) = u + iv$ in terms of z . 6
- C) Evaluate $\oint_C \frac{e^{-z}}{z+1} dz$ where C is the circle
 i) $|z| = 2$
 ii) $|z| = \frac{1}{2}$ 6

*** End ***